Week 9: Representation theory and lattice models

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Our goal today is to use representation theory to produce R-matrices, and hence lattice models. The construction of Drinfeld and Jimbo gives a general way to do this using quantum groups.

1 Representation theory

Recall that in the six-vertex model, we attach the vector space

$$V = \mathbb{C}^2 = \text{span}\{\uparrow, \downarrow\}$$

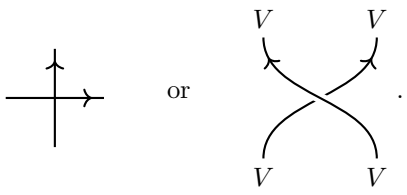
to each line. In the ordered basis

$$\uparrow\uparrow, \quad \uparrow\downarrow, \quad \downarrow\uparrow, \quad \downarrow\downarrow$$

of $V \otimes V$, the local weights give the matrix

$$R = \begin{pmatrix} a_1 & 0 & 0 & 0 \\ 0 & b_2 & c_1 & 0 \\ 0 & c_2 & b_1 & 0 \\ 0 & 0 & 0 & a_2 \end{pmatrix}.$$

We may draw a crossing of two lines as



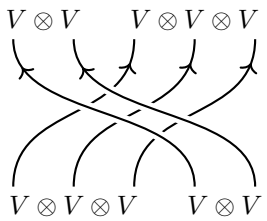
The second picture swaps the two tensor factors. If $P(v \otimes w) = w \otimes v$, then its operator is $\check{R} = PR$.

1.1 Braided tensor categories

The natural setting for these crossings is a **braided monoidal category**. In such a category, there are natural isomorphisms

$$c_{U,W} : U \otimes W \xrightarrow{\sim} W \otimes U$$

compatible with the tensor product. The Yang-Baxter equation is a consequence of the axioms. For example, we can cross two blocks of tensor factors:



There is a corresponding algebraic picture. An algebra A gives a linear category A-mod, and a Hopf algebra gives a tensor product on its category of representations. A quasitriangular structure then gives a braiding. We can organize these constructions as follows:

$$\text{linear category} \xleftarrow{\text{modules}} \text{algebra}$$

$$\begin{array}{ccc} \text{tensor category} & \longleftarrow & \text{Hopf algebra} \\ \Big\Downarrow{\scriptstyle \text{center}} & & \Big\Downarrow{\scriptstyle \text{Drinfeld double}} \\ \text{braided tensor category} & \longleftarrow & \text{quasitriangular Hopf algebra} \end{array}$$

The Drinfeld double is the algebraic counterpart of passing to the center.

Definition 1.1. The **Drinfeld center** $Z(\mathcal{C})$ has objects (V, c), where $V \in \mathcal{C}$ and

$$c_X : V \otimes X \xrightarrow{\sim} X \otimes V$$

is natural in X and compatible with tensor products. Suppressing associativity isomorphisms, this means

$$c_{X\otimes Y} = (\mathrm{Id}_X \otimes c_Y)(c_X \otimes \mathrm{Id}_Y).$$

The crossing with the tensor unit is the identity.

This leads us to quantum groups.

1.2 $U_q(\mathfrak{sl}_2)$

Recall that $\mathfrak{sl}_2$ is generated by H, E, F with relations

$$[H, E] = 2E, \qquad [H, F] = -2F, \qquad [E, F] = H.$$

Its natural representation is given by

$$H = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \qquad E = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \qquad F = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}.$$

The quantum group $U_q(\mathfrak{sl}_2)$ is generated by $K^{\pm 1}, E, F$ with relations

$$KEK^{-1} = q^2 E, \qquad KFK^{-1} = q^{-2} F, \qquad [E, F] = \frac{K - K^{-1}}{q - q^{-1}}.$$

Here we take $q \in \mathbb{C}^\times$ to be generic. To compare with $\mathfrak{sl}_2$, write $K = q^H$. As $q \to 1$,

$$\frac{q^H - q^{-H}}{q - q^{-1}} \longrightarrow H.$$

1.3 The universal R-matrix

For a quantum group $U_q(\mathfrak{g})$, we can find a **universal R-matrix** $\mathcal{R}$. It lies in a completion of the tensor product of the two Borel subalgebras:

$$\mathcal{R} \in U_q(\mathfrak{b}^-) \widehat{\otimes} U_q(\mathfrak{b}^+), \qquad U_q(\mathfrak{b}^-) = \langle K_i^{\pm 1}, F_i \rangle, \quad U_q(\mathfrak{b}^+) = \langle K_i^{\pm 1}, E_i \rangle.$$

Let Δ be the coproduct, and let $\Delta^{\mathrm{op}} = P \circ \Delta$ be the coproduct with its factors swapped. The universal R-matrix satisfies

$$\mathcal{R}\Delta(a) = \Delta^{\mathrm{op}}(a)\mathcal{R}.$$

Evaluating on representations gives an operator $R_{U,W}$ on $U \otimes W$. If $\rho = \rho_U \otimes \rho_W$, then the following diagram commutes:

$$\begin{array}{ccc} U \otimes W & \xrightarrow{R_{U,W}} & U \otimes W \\ {\scriptstyle \rho(\Delta(a))}\downarrow & & \downarrow{\scriptstyle \rho(\Delta^{\mathrm{op}}(a))} \\ U \otimes W & \xrightarrow[R_{U,W}]{} & U \otimes W \end{array}$$

Thus $PR_{U,W}$ is a map of representations from $U \otimes W$ to $W \otimes U$.

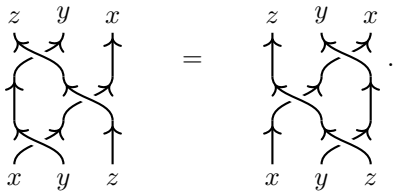
Example 1.2. For $\mathfrak{g} = \mathfrak{sl}_2$, let

$$V = \mathbb{C}(q) \uparrow \oplus \mathbb{C}(q) \downarrow$$

be the quantum version of the natural representation. Up to an overall scalar, the universal R-matrix acts on $V \otimes V$ by

$$R = \begin{pmatrix} q & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & q - q^{-1} & 1 & 0 \\ 0 & 0 & 0 & q \end{pmatrix}.$$

The crossings satisfy the braid relation:



In terms of the operators without the permutation, this is

$$R_{12}R_{13}R_{23} = R_{23}R_{13}R_{12}.$$

2 Spectral parameters and loop algebras

We want R-matrices with spectral parameters. For this, we pass to affine Lie algebras and affine quantum groups, or to their loop versions.

For a semisimple Lie algebra $\mathfrak{g}$, the root system, Cartan matrix, or Dynkin diagram determines $\mathfrak{g}$ and $U_q(\mathfrak{g})$. More generally, starting from a generalized Cartan matrix gives a Kac-Moody algebra and a quantum Kac-Moody algebra.

In untwisted affine type, omitting the degree derivation gives a central extension of the **loop algebra**

$$L\mathfrak{g} = \mathfrak{g}[z^{\pm 1}] = \mathfrak{g} \otimes \mathbb{C}[z^{\pm 1}].$$

Here we work with the loop algebra, with bracket

$$[X \otimes z^n, Y \otimes z^m] = [X, Y] \otimes z^{n+m}.$$

There is a similar construction for quantum groups.

Example 2.1. The affine diagram for $\mathfrak{sl}_2$ has two vertices:



$$\begin{pmatrix} 2 & -2 \\ -2 & 2 \end{pmatrix}.$$

2.1 Two presentations

There are two useful presentations of $U_q(L\mathfrak{g})$. The **Serre presentation** uses generators

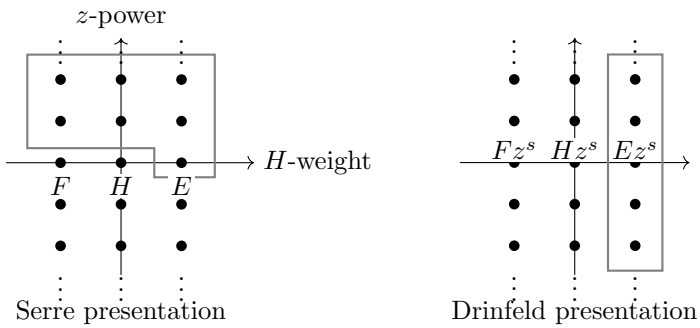
$$K_i^{\pm 1}, E_i, F_i, \qquad i \in I_{\text{aff}}.$$

The **Drinfeld presentation** uses loop generators

$$K_i^{\pm 1}, \quad H_{i,r}, \quad E_{i,s}, \quad F_{i,s}, \qquad i \in I, \quad r \in \mathbb{Z} \setminus \{0\}, \quad s \in \mathbb{Z}.$$

Here I is the set of vertices of the finite Dynkin diagram. In the classical limit, $E_{i,s}$ corresponds to $E_i \otimes z^s$.

These presentations give two different triangular decompositions. For $L\mathfrak{sl}_2$, the two pictures look like:



In the first picture, the positive part consists of the positive powers of z, together with E at power zero. The negative part consists of the negative powers, together with F at power zero. The Cartan part is $\mathbb{C}H$. In the second picture, we use a different positive part, consisting of all Ez^s in the right column for $s \in \mathbb{Z}$. The Cartan part consists of all Hz^s in the middle column.

The Serre presentation is useful for integrable representations and highest weight module theory. The Drinfeld presentation is useful for finite dimensional representations.

3 Finite dimensional representations

We can think of the loop algebra as algebraic maps:

$$L\mathfrak{g} = \operatorname{Map}(\mathbb{C}^\times, \mathfrak{g}).$$

Given a $\mathfrak{g}$-representation V and a point $a \in \mathbb{C}^\times$, we obtain an **evaluation representation** by

$$(X \otimes f(z))v = f(a)Xv.$$

Thus we can place $\mathfrak{g}$-representations at finitely many points of

$$\mathbb{C}^\times = \operatorname{Spec} \mathbb{C}[z^{\pm 1}].$$

For example, if V is a $\mathfrak{g}$-representation, then

$$V \otimes \mathbb{C}[z^{\pm 1}]/(z - a)$$

is supported at the point a. Tensoring irreducible evaluation representations at distinct points gives all the finite dimensional irreducible representations of $L\mathfrak{g}$.

The categories $\operatorname{Rep}(L\mathfrak{g})$ and $\operatorname{Rep}(U_q(L\mathfrak{g}))$ are not semisimple. Both also have a rotation symmetry. Classically, rotation sends

$$X \otimes z^n \longmapsto a^n X \otimes z^n.$$

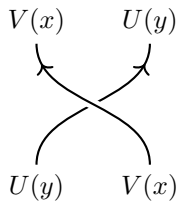
The quantum loop algebra has a corresponding automorphism τ_a. If V is a representation, write $V(a)$ for the same vector space with its action twisted by τ_a.

3.1 The R-matrix with spectral parameters

The category of finite dimensional quantum loop representations is not a braided category. The R-matrix can have poles, or it can fail to be invertible at a special value of the parameters. We nevertheless obtain a map

$$\check{R}_{U,V}(y,x) : U(y) \otimes V(x) \longrightarrow V(x) \otimes U(y)$$

for generic x, y.



Example 3.1. Take $U = V = \mathbb{C}^2$ for $U_q(L\mathfrak{sl}_2)$. In the basis

$$\uparrow\uparrow, \quad \uparrow\downarrow, \quad \downarrow\uparrow, \quad \downarrow\downarrow,$$

the normalized crossing is

$$\check{R}_{U,V}(y,x) = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & \dfrac{(q^2-1)x}{q^2x-y} & \dfrac{q(x-y)}{q^2x-y} & 0 \\ 0 & \dfrac{q(x-y)}{q^2x-y} & \dfrac{(q^2-1)y}{q^2x-y} & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}.$$

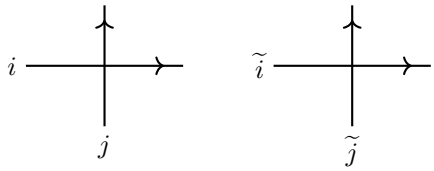
This is not defined when $y = q^2x$. Its determinant is

$$\det \check{R}_{U,V}(y,x) = \frac{q^2y - x}{q^2x - y},$$

so it is singular when $x = q^2y$.

Theorem 3.2 (Drinfeld)**.** *For finite dimensional irreducible quantum loop representations, the R-matrix, after scalar normalization, is a rational function of the ratio x/y.*

The same construction works with other quantum groups, giving analogs of the six-vertex model. For example, we can use $U_q(\mathfrak{sl}(1|1))$ in place of $U_q(\mathfrak{sl}_2)$. We label the lines by representations, and obtain vertex weights by taking matrix coefficients of the R-matrix:



The choice of quantum group and representations determines the local states and their weights.

AI usage

AI was used for writing these notes based on photos of the blackboard during the talk.